

PROJECT PHASE 4: MODEL IMPROVEMENT, OPTIMIZATION AND COMPARATIVE EXPERIMENTS

Deadline: Week 9

Weight: 10% of the project grade

Penalty: Each delay is penalized with 3 points

Maximum allowed delays: 2

Objective

The goal of this phase is to improve the baseline model developed in Phase 3 by testing a stronger or alternative modelling approach and performing systematic experiments.

Students must go beyond the first working model and investigate whether the performance can be improved through:

- a different algorithm
- hyperparameter tuning
- better modelling choices
- structured comparison between models

This phase is focused on model improvement and experimental validation.

The purpose is not only to obtain better performance, but also to understand why one approach works better than another for the selected problem and dataset. Phase 3 established the baseline model and initial evaluation, including error analysis that should guide the improvements in this phase.

Coding (Improved Modelling & Experiments)

In this phase, you must extend your work from Phase 3 and build an improved modelling pipeline.

1. Improved Model

- Develop an improved version of the modelling approach from Phase 3
- The improvement must be meaningful and clearly justified
- Examples:
 - better feature usage
 - better preprocessing for modelling

- more suitable model architecture
- improved handling of class imbalance / rare cases / noise

2. Alternative Algorithm

- You must implement at least one additional algorithm different from the baseline model
- The second algorithm must be relevant to the problem type
- Examples:
 - classification: Logistic Regression → Random Forest / XGBoost / SVM / neural network
 - regression: Linear Regression → Random Forest Regressor / XGBoost / Gradient Boosting
 - clustering / pattern discovery: K-Means → DBSCAN / Hierarchical Clustering / Gaussian Mixture Models
 - anomaly detection: Isolation Forest → One-Class SVM / Local Outlier Factor / Autoencoder

You must justify why the new algorithm is suitable for your problem.

3. Hyperparameter Tuning

- Perform hyperparameter tuning for at least one of the models
- You may use:
 - grid search
 - random search
 - manual but well-structured tuning
- You must clearly specify:
 - which hyperparameters were tested
 - the tested values / search space
 - the selection criterion
 - the final chosen configuration

The tuning process must be explained, not only executed.

4. Model Comparison

- Compare the models using appropriate evaluation metrics depending on the problem
- The comparison must include:
 - quantitative results (tables, plots)
 - interpretation of differences
 - discussion of trade-offs

Examples of trade-offs:

- better accuracy but slower training
- better recall but lower precision
- better regression fit but higher sensitivity to outliers
- better clustering quality but lower interpretability

5. Discussion of Improvements

You must analyze:

- whether the new model actually improves the baseline
- where the improvement appears
- where performance remains weak
- possible reasons for improvement or lack of improvement

This section is very important: students must not only report numbers, but explain the experimental results in a critical way.

Report

Please continue to work on the report template provided:

<https://github.com/SergiuLimboi/Intelligent-techniques-for-processing-structured-and-large-data/tree/main/Report%20template>

As already specified in the earlier project phases, students must continue building the same cumulative report rather than creating a separate report for each phase.

IMPORTANT:

- You do NOT create separate reports for each phase
- This phase, and all phases, must be integrated into the SAME report
- Conclusions, introduction, and abstract are written at the end of the project

The report must include:

1. Improved Modelling Approaches

- Description of the improved model(s)
- Description of the additional algorithm
- Motivation for the new approaches
- Why these approaches were selected
- Link with the observations from Phase 3 error analysis

2. Hyperparameter Tuning Strategy

- What was tuned
- Search strategy used
- Final selected hyperparameters
- Justification of the final choice

3. Comparison Between Models

- Clear comparison between:
 - baseline model
 - improved model
 - alternative model
- Must include:
 - tables and/or plots
 - interpretation of differences
 - discussion of strengths and weaknesses of each model

4. Experimental Findings

- What improved and what did not
- Which model is more appropriate for the problem
- What the experiments reveal about the dataset and modelling choices

5. Clarity and Structure

- The section must be clearly written and easy to follow
- Results must be explained, not only presented

- Visual elements (tables, plots, diagrams) should support the discussion

Deliverables (Mandatory)

Report:

- **LaTeX source**
- **PDF version**

Code:

- **Jupyter Notebook / Python scripts**
- **Must be runnable**
- **Must include explanations**

Repository:

- **GitHub Classroom (same repo)**

Submission rule:

- **Upload before laboratory in Week 9**
- **Do not push only an archive!**
- **Push the deliverables on the main branch !**

Grading

- **Report: 50%**
- **Implementation (code): 50%**

Final grade = weighted average

Grades will be provided after evaluation by course & lab instructors, following the same grading logic used in the previous phases.

Grading Rubric — Code (Improved Modelling & Experiments)

Section	Points	Full points	Partial points	No points
1 . Improved model	3	Improved model is meaningful, well-implemented, and clearly justified	Improvement attempted but weak or insufficiently justified	Missing
2. Alternative algorithm	2.5	A relevant second algorithm is implemented and appropriately explained	Second algorithm exists but is weakly motivated or poorly explained	Missing
3. Hyperparameter tuning	2	Tuning is structured, relevant, and clearly documented	Tuning attempted but superficial or unclear	Missing
4 . Model comparison	1.5	Correct metrics, clear comparison, and interpretation	Metrics shown but little interpretation	Missing
5 . Code clarity	1	Clean, documented	Superficial	Poor
Total	10 points			

Grading Rubric — Report Phase 4

Section	Points	Full points	Partial points	No points
1 . Improved modelling approaches	2	New approaches are clearly described and justified	Basic description without strong justification	Missing
2 . Hyperparameter tuning strategy	2	Tuning process is clearly explained and justified	Tuning mentioned but poorly explained	Missing
3 . Comparison between models	3	Strong comparative analysis with interpretation and evidence	Comparison present but shallow	Missing
4 . Experimental findings	2	Insightful discussion of improvements, limitations, and lessons learned	Some observations, but limited insight	Missing
5 . Clarity & structure	1	Well written, structured	Acceptable	Poor
Total	10 points			